

Data-Based Iterative Dynamic Decoupling Control for Precision MIMO Motion Systems

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Abstract—Decoupling control is still an important research topic for precision multiple-input-multiple-output (MIMO) motion systems involved in computer numerically controlled (CNC) machine tools, wafer scanners, etc. In this paper, to minimize internal coupling and improve servo performance, a data-based iterative dynamic decoupling control (IDDC) approach is synthesized. Specifically, a MIMO dynamic decoupling controller structured with finite impulse response filter is used as an add-on to a static decoupling part. Then, a data-based parameter optimization algorithm is developed such that the optimal parameters can be iteratively solved based entirely on the input/output data by minimizing the coupling-induced error. Unlike pre-existing IDDC approaches, the proposed approach can achieve an unbiased estimate of the optimal parameters combined with a small estimate variance that is illustrated through numerical simulation. Finally, application to an ultraprecision wafer stage confirms that the proposed approach significantly decreases the coupling-induced error and achieves enhanced performance compared to pre-existing approaches.

Index Terms—Dynamic decoupling, data-based, iterative optimization, multiple-input-multiple-output (MIMO), precision motion systems.

I. INTRODUCTION

MODERN industrial equipment, such as computer numerically controlled (CNC) machine tools [1], [2], wafer scanners [3], and hard disks [4], [5], generally requires multiple degree-of-freedom (DOF) motion with high accuracy to implement manufacturing and processing. Therefore, the underlying motion systems are clearly multiple-input-multiple-output

(MIMO). For example, the state-of-the-art wafer scanners employ a photolithographic process to manufacture integrated circuits (ICs) on a silicon wafer. During this exposure process, the wafer stage holds the wafer and accurately tracks a predefined reference trajectory in six motion DOF. For MIMO systems, internal coupling is one of the main obstacles to high servo performance [6], [7]. Consequently, the decoupling control design for precision MIMO motion systems is still an important research topic.

A static decoupling control design is widely employed to decrease the internal coupling of MIMO motion systems. On the basis of the resulting diagonal dominant plant, a set of single-input-single-output (SISO) controllers are separately designed to achieve the specific performance of each motion DOF [8], [9]. This control method for MIMO systems is simple to be implemented, and preserves the SISO control design along with its advantages, as discussed in [10]. Due to the increasing demands on system performance and production quality, industrial motion systems typically exhibit high-order dynamics related to flexible modes. As a result, it is getting harder for the static decoupling control to achieve satisfactory decoupling.

In order to improve the performance of MIMO motion systems, multivariable control can be employed to replace the SISO control part [11], [12]. However, it is quite complex to design the multivariable controllers, and the multi-intersected feedback paths may lead to internal uncertainties. On-line estimate of the interactive dynamics based on the observer [13]–[15] or neural-network [16], [17] can also be exploited to tolerate the residual coupling. Nonetheless, due to the inevitable estimate error and a relatively long estimate process, the application of on-line estimators to precision MIMO motion systems generally leads to a poor response and even instability before reaching the steady state.

A more natural approach to further decrease the internal coupling of MIMO motion systems is to attach a MIMO dynamic decoupling part to the static decoupling part [18], [19] and leave the SISO controllers intact. Generally, the dynamic decoupling control (DDC) design is based on the physical model from first principles. Reliable models of industrial MIMO systems are often difficult or impossible to obtain due to the high complexity of plants, the excessive cost of modeling, and even machine-to-machine variation. On the other hand, the inevitable modeling error probably leads to unsatisfactory performance when the model-based DDC is applied to real plants.

In contrast to model-based approaches, the usage and interpretation of input/output (I/O) data have become attractive

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topics for industrial engineers [20]–[22]. Therefore, data-based control approaches, including virtual reference feedback tuning (VRFT) [23], [24], correlation-based tuning (CbT) [25], [26], and iterative feedback tuning (IFT) [27], [28], are significantly explored for performance improvement. Data-based approaches avoid the need for modeling and make direct use of the measured I/O data to optimize the parameters of a controller that has a predefined fixed structure [29]. IFT requires no plant model and can achieve the performance optimization under the disturbance situation at hand. Hence, IFT seems appealing for the feedback control design, and has been extended to feedforward tuning [30], [31]. IFT has been successfully applied in many industrial applications, including robotics [32], wafer stages [9], [33], etc. In [34], IFT is exploited to tune a DDC controller. Since the error gradient is estimated via parameter perturbation method, this approach results in a biased estimate of the optimal parameters combined with a large estimate variance.

In this paper, to minimize the internal coupling and improve the servo performance, a data-based iterative DDC (IDDC) approach inspired by IFT is significantly synthesized for precision MIMO motion systems. Specifically, a MIMO DDC controller structured with finite impulse response (FIR) filter is used as an add-on to the static decoupling part. The performance criterion is defined as the quadratic function w.r.t. the coupling-induced error. An unbiased estimate method for the gradient of the performance criterion is developed based entirely on I/O data of the closed-loop MIMO system. Since the plant incorporating the static decoupling part is diagonal dominant, it is believed that the optimal parameters of the DDC controller are close to zero. Hence, starting from zero, gradient-based Gauss–Newton algorithm (GNA) is utilized to iteratively tune the parameters. Owing to the unbiasedness of the estimated gradient of the performance criterion and the nearness of the optimal parameters to zero, a fast convergence can be guaranteed by the proposed approach. Throughout the parameter tuning process, the DDC controller is inherently stable, and no model or system knowledge is needed. The novelty of this paper is threefold. First, DDC structured with FIR filter is proposed for precision MIMO motion systems. Second, a novel iterative parameter optimization algorithm is developed such that the optimal parameters of the DDC controller can be unbiasedly solved based on the I/O data. Third, numerical simulation and application to an ultra-precision wafer stage illustrate the effectiveness of the proposed approach.

The rest of this paper is organized as follows. In Section II, the problem formulation is stated. Then, in Section III, a novel data-based IDDC approach is proposed. In Section IV, comparative simulations and experiments are presented to illustrate the effectiveness of the proposed approach. Finally, Section V concludes this paper.

II. PROBLEM STATEMENT

As improved accuracy combined with high speed is required, decoupling control is still a main concern for precision MIMO motion systems.

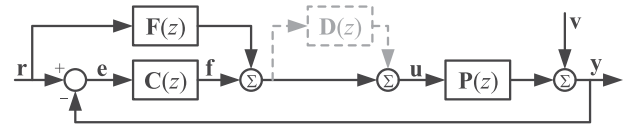


Fig. 1. Block diagram of a typical control configuration for precision MIMO motion systems.

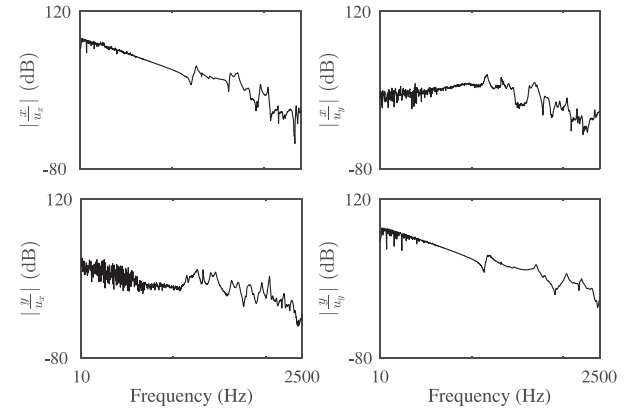


Fig. 2. Magnitude–frequency characteristics corresponding to the x -axis and y -axis of a MIMO wafer stage.

A. Nominal Control Design

A typical control configuration for precision MIMO motion systems is shown in Fig. 1. $P(z) \in \mathbb{C}^{n \times n}$ denotes a MIMO plant with n being the number of DOF. Incorporating a static and model-based actuator/sensor decoupling part, $P(z)$ is diagonal dominant. $C(z) \in \mathbb{C}^{n \times n}$ and $F(z) \in \mathbb{C}^{n \times n}$ denote the diagonal feedback and the diagonal feedforward controller, respectively. The entries of $C(z)$ and $F(z)$ are designed via SISO design methods. $C(z)$ robustly stabilizes the closed-loop system and enhances the disturbance rejection ability. $F(z)$ aims at decreasing the reference-induced servo error. Given the reference $r \in \mathbb{R}^n$ and the output $y \in \mathbb{R}^n$, the servo error $e \in \mathbb{R}^n$ is constructed as $e = r - y$. f , u , and v denote the output of $C(z)$, the plant input, and the immeasurable disturbance, respectively. Assume that the element $v_i (i = 1, \dots, n)$ of v is given by $v_i = K_i(z)\varepsilon_i$ where $K_i(z)$ is monic, and ε_i is white noise with zero mean.

In general, the static decoupling part in $P(z)$ could not fully achieve the desired decoupling of precision motion systems due to high-order dynamics related to flexible modes. Fig. 2 depicts the magnitude–frequency characteristics corresponding to the x -axis and y -axis of a MIMO wafer stage [31]. Though diagonal dominant, the residual coupling of $P(z)$ leads to performance limitations and cannot be ignored as the nanopositioning wafer stage is subject to increasing performance requirements.

B. Dynamic Decoupling Control

To further decrease the internal coupling and improve the servo performance, a DDC controller $D(z) \in \mathbb{C}^{n \times n}$ is added to

		M_1	M_2	M_3
$\mathbf{r} = [r_b, 0]^T$	$\frac{1}{2N} \ e_2\ ^2 \text{ (nm}^2\text{)}$	20.7	11.0	6.5
	$\max(e_2) \text{ (nm)}$	29	22	13
$\mathbf{r} = [0, r_b]^T$	$\frac{1}{2N} \ e_1\ ^2 \text{ (nm}^2\text{)}$	52.5	31.3	24.8
	$\max(e_1) \text{ (nm)}$	40	30	31

Fig. 3. DDC system with $n = 2$.

the nominal control system, as shown in Fig. 1, and given by

$$\mathbf{D}(z) = \begin{bmatrix} 0 & D_{12}(z) & \cdots & D_{1n}(z) \\ D_{21}(z) & 0 & \cdots & D_{2n}(z) \\ \vdots & \vdots & \ddots & \vdots \\ D_{n1}(z) & D_{n2}(z) & \cdots & 0 \end{bmatrix}. \quad (1)$$

The addition of $\mathbf{D}(z)$ does not change the original control configuration, and the SISO control design of $\mathbf{C}(z)$ and $\mathbf{F}(z)$, along with its advantages, can be preserved.

$\mathbf{F}(z)$ is generally designed after $\mathbf{D}(z)$. Hence, set $\mathbf{F}(z) = \mathbf{0}$ in the following discussion on the design of $\mathbf{D}(z)$. For simplicity, $n = 2$ is considered in the remainder of this section. Fig. 3 shows the block diagram of the DDC system with $n = 2$. Then, we have

$$\mathbf{P}(\mathbf{I} + \mathbf{D}) = \begin{bmatrix} P_{11} + P_{12}D_{21} & P_{12} + P_{11}D_{12} \\ P_{21} + P_{22}D_{21} & P_{22} + P_{21}D_{12} \end{bmatrix}. \quad (2)$$

Ideally, the DDC design is to make the off-diagonal elements of $\mathbf{P}(\mathbf{I} + \mathbf{D})$ zero. Consequently, it should be satisfied that

$$\begin{cases} D_{12} = -P_{12}/P_{11} \\ D_{21} = -P_{21}/P_{22} \end{cases}. \quad (3)$$

Through (3), the accurate dynamical model $\mathbf{P}(z)$ is required if the model-based method is employed to construct $D_{ij}(z)$. However, in practice, the modeling error is inevitable, especially for motion systems with complex dynamics. Furthermore, the dynamical model does not address the uncertainty and machine-to-machine variation encountered in the precision motion industry. Consequently, the servo performance of the traditional model-based DDC employing (3) is limited.

III. PROPOSED DATA-BASED IDDC APPROACH

For precision industrial motion systems, data-based approaches provide a potential way to avoid modeling error and address the machine-specific dynamic differences. Therefore, a data-based IDDC approach inspired by IFT is proposed to achieve the parameter optimization of $\mathbf{D}(z)$ through minimizing the coupling-induced error.

Definition 1: The off-diagonal element $D_{ij}(z)$ ($j \neq i$) of $\mathbf{D}(z)$ is linearly parameterized as

$$\mathcal{D} = \left\{ D_{ij}(\boldsymbol{\theta}_{ij}, z) \mid D_{ij}(\boldsymbol{\theta}_{ij}, z) = \boldsymbol{\Psi} \boldsymbol{\theta}_{ij} = \sum_{m=0}^{\kappa} \theta_{ij,m} z^{-m} \right\} \quad (4)$$

with the order κ , parameters $\boldsymbol{\theta}_{ij} = [\theta_{ij,0}, \theta_{ij,1}, \dots, \theta_{ij,\kappa}]^T \in \mathbb{R}^{\kappa+1}$, and basis functions $\boldsymbol{\Psi} = [1, z^{-1}, \dots, z^{-\kappa}]$.

$D_{ij}(\boldsymbol{\theta}_{ij}, z)$ is structured with FIR filter, and its poles are all located at the origin. Hence, $D_{ij}(\boldsymbol{\theta}_{ij}, z)$ is inherently stable regardless of $\boldsymbol{\theta}_{ij}$. Denote the desired DDC controller as $\mathbf{D}^*(z)$ that satisfies $\mathbf{P}(\mathbf{I} + \mathbf{D}^*) = \mathbf{0}$. It is well known that a causal and stable system can be well described by a finite-order model structured with FIR filter. If the off-diagonal elements of $\mathbf{D}^*(z)$ are stable, it can be believed that $\mathbf{D}(z)$ defined in (1) with $D_{ij}(z)$ defined in (4) possesses the ability to diagonalize $\mathbf{P}(\mathbf{I} + \mathbf{D})$ when κ is large enough.

A. Data-Based Iterative Parameter Optimization

When only the reference of the i th DOF, i.e. r_i , is excited, the errors $e_1, \dots, e_{i-1}, e_{i+1}, \dots, e_n$ are mainly induced by the cross coupling from the i th DOF to other DOFs, and can be compensated for by $D_{1i}(z), \dots, D_{(i-1)i}(z), D_{(i+1)i}(z), \dots, D_{ni}(z)$. In this paper, sequential excitation is adopted in the data-based parameter optimization of $\mathbf{D}(z)$. Specifically, when optimizing for the i th DOF, only r_i is persistently excited, i.e., $\mathbf{r}_i = [\underbrace{0, \dots, 0}_{(i-1) \text{ zeros}}, r_i, \underbrace{0, \dots, 0}_{(n-i) \text{ zeros}}]^T$ is employed as

the reference of the closed-loop system. Accordingly, parameters $\boldsymbol{\theta}_i = [\boldsymbol{\theta}_{1i}^T, \dots, \boldsymbol{\theta}_{(i-1)i}^T, \boldsymbol{\theta}_{(i+1)i}^T, \dots, \boldsymbol{\theta}_{ni}^T]^T$ are to be optimized so as to minimize the coupling-induced error $\mathbf{e}_i = [e_1, \dots, e_{i-1}, e_{i+1}, \dots, e_n]^T \in \mathbb{R}^{n-1}$.

Definition 2: The performance criterion of the proposed data-based IDDC for the i th DOF is given by

$$J_i(\boldsymbol{\theta}_i) = \frac{1}{2N} \mathbb{E} \left[\sum_{t=1}^N \mathbf{e}_i^T(t, \boldsymbol{\theta}_i) \mathbf{Q}_i \mathbf{e}_i(t, \boldsymbol{\theta}_i) \right] \quad (5)$$

where t denotes the sampling index, N denotes the sampling number, and the diagonal positive definite matrix $\mathbf{Q}_i \in \mathbb{R}^{(n-1) \times (n-1)}$ is a user-defined weighting matrix to emphasize the performance of certain DOF. $\mathbb{E}[\cdot]$ denotes expectation w.r.t. the disturbance $\mathbf{v}$. The optimal parameters $\boldsymbol{\theta}_i^*$ are defined by

$$\boldsymbol{\theta}_i^* = \arg \min_{\boldsymbol{\theta}_i} J_i(\boldsymbol{\theta}_i). \quad (6)$$

Since $\mathbf{P}(z)$, incorporating a static decoupling part, is diagonal dominant, it can be believed that $\boldsymbol{\theta}_i^*$ are close to zeros. Gradient-based GNA is employed to iteratively solve $\boldsymbol{\theta}_i^*$

$$\boldsymbol{\theta}_i^{k+1} = \boldsymbol{\theta}_i^k - \gamma_i^k (\mathbf{H}_i^k)^{-1} \frac{\partial J_i}{\partial \boldsymbol{\theta}_i}(\boldsymbol{\theta}_i^k) \quad (7)$$

with the iteration index k and the initial parameters $\boldsymbol{\theta}_i^0 = \mathbf{0}$. From (5), the gradient $\partial J_i / \partial \boldsymbol{\theta}_i$ is given by

$$\frac{\partial J_i}{\partial \boldsymbol{\theta}_i}(\boldsymbol{\theta}_i^k) = \frac{1}{N} \mathbb{E} \left[\sum_{t=1}^N \frac{\partial \mathbf{e}_i^T}{\partial \boldsymbol{\theta}_i}(t, \boldsymbol{\theta}_i^k) \mathbf{Q}_i \mathbf{e}_i(t, \boldsymbol{\theta}_i^k) \right]. \quad (8)$$

$\mathbf{H}_i$ is a positive Gauss-Newton approximation to the Hessian of $J_i(\boldsymbol{\theta}_i)$ that enables to speed up the convergence rate, and

given by

$$\mathbf{H}_i^k = \frac{1}{N} \mathbb{E} \left[\sum_{t=1}^N \frac{\partial \mathbf{e}_i^T}{\partial \boldsymbol{\theta}_i}(t, \boldsymbol{\theta}_i^k) \mathbf{Q}_i \frac{\partial \mathbf{e}_i}{\partial \boldsymbol{\theta}_i}(t, \boldsymbol{\theta}_i^k) \right]. \quad (9)$$

The step size $0 < \gamma_i^k \leq 1$ should be selected to balance the tradeoff between the convergence rate and convergence stability. When $\boldsymbol{\theta}_i^*$ for all DOFs are iteratively solved by (7), the optimal DDC controller $\mathbf{D}(\boldsymbol{\theta}^*, z)$ with $\boldsymbol{\theta}^* = [(\boldsymbol{\theta}_1^*)^T, \dots, (\boldsymbol{\theta}_n^*)^T]^T$ is obtained.

B. Estimate Method for the Gradient and Hessian

From (8) and (9), it is clear that finding $\boldsymbol{\theta}_i^*$ in (6) with (7) requires the error gradient $\partial \mathbf{e}_i / \partial \boldsymbol{\theta}_i$ at each iteration k . From Fig. 1, the servo error $\mathbf{e}$ can be expressed as

$$\mathbf{e} = \mathbf{S}(\mathbf{r} - \mathbf{v}) \quad (10)$$

with sensitivity function $\mathbf{S} = [\mathbf{I} + \mathbf{P}(\mathbf{I} + \mathbf{D})\mathbf{C}]^{-1}$ and identity matrix $\mathbf{I}$.

For an invertible matrix $\mathbf{A}$, the partial derivative of $\mathbf{A}^{-1}$ w.r.t. a scalar x can be written as [35]

$$\frac{\partial \mathbf{A}^{-1}}{\partial x} = -\mathbf{A}^{-1} \frac{\partial \mathbf{A}}{\partial x} \mathbf{A}^{-1}. \quad (11)$$

Therefore, for the first parameter $\theta_{ji,0}$ of $D_{ji}(z)$ ($j \neq i$), we have from (11) that

$$\frac{\partial \mathbf{S}}{\partial \theta_{ji,0}} = -\mathbf{S} \mathbf{P} \frac{\partial \mathbf{D}}{\partial \theta_{ji,0}} \mathbf{C} \mathbf{S}. \quad (12)$$

Combining (10) and (12) yields

$$\frac{\partial \mathbf{e}}{\partial \theta_{ji,0}} = -\mathbf{S} \mathbf{P} \mathbf{w}_{ji} \quad (13)$$

with $\mathbf{w}_{ji} = \frac{\partial \mathbf{D}}{\partial \theta_{ji,0}} \mathbf{f}$ and $\mathbf{f} = \mathbf{C} \mathbf{e}$. Considering the definition of $\mathbf{D}(z)$ in (1) and $D_{ji}(z)$ in (4), $\mathbf{w}_{ji}$ can be rewritten as

$$\mathbf{w}_{ji} = \left[\underbrace{0, \dots, 0}_{(j-1) \text{ zeros}}, f_i, \underbrace{0, \dots, 0}_{(n-j) \text{ zeros}} \right]^T. \quad (14)$$

Equations (13) and (14) lead to the following basic method for the estimate of the error gradient $\frac{\partial \mathbf{e}_i}{\partial \theta_{ji,0}}(\boldsymbol{\theta}_i^k)$ through two experiments with $\boldsymbol{\theta}_i = \boldsymbol{\theta}_i^k$.

1) Tracking Experiment I: Excite only r_i , i.e., $\mathbf{r} = \mathbf{r}_i$, and record $\mathbf{e}^I$ and $\mathbf{f}^I$ (the superscript I refers to the experiment number). Construct

$$\mathbf{w}_{ji}^I = \left[\underbrace{0, \dots, 0}_{(j-1) \text{ zeros}}, f_i^I, \underbrace{0, \dots, 0}_{(n-j) \text{ zeros}} \right]^T. \quad (15)$$

2) Gradient Experiment II: Set $\mathbf{r} = \mathbf{0}$, inject $k_{ji} \mathbf{w}_{ji}^I$ (the gain $k_{ji} > 0$) as the plant input $\mathbf{u}$, and record $\mathbf{e}^{II}$. It follows that

$$\mathbf{e}^{II} = -k_{ji} \mathbf{S} \mathbf{P} \mathbf{w}_{ji}^I - \mathbf{S} \mathbf{v}^{II}. \quad (16)$$

Hence, from (13), an estimate of $\frac{\partial \mathbf{e}}{\partial \theta_{ji,0}}(\boldsymbol{\theta}_i^k)$ can be given by

$$\widehat{\frac{\partial \mathbf{e}}{\partial \theta_{ji,0}}}(\boldsymbol{\theta}_i^k) = \frac{1}{k_{ji}} \mathbf{e}^{II}. \quad (17)$$

Accordingly, the estimate of $\frac{\partial \mathbf{e}_i}{\partial \theta_{ji,0}}(\boldsymbol{\theta}_i^k)$ is given by

$$\widehat{\frac{\partial \mathbf{e}_i}{\partial \theta_{ji,0}}}(\boldsymbol{\theta}_i^k) = \frac{1}{k_{ji}} [e_1^{II}, \dots, e_{i-1}^{II}, e_{i+1}^{II}, \dots, e_n^{II}]^T. \quad (18)$$

Owing to the FIR structure of $D_{ji}(z)$, it follows that

$$\widehat{\frac{\partial \mathbf{e}_i}{\partial \theta_{ji,m}}}(\boldsymbol{\theta}_i^k) = z^{-m} \widehat{\frac{\partial \mathbf{e}_i}{\partial \theta_{ji,0}}}(\boldsymbol{\theta}_i^k), m = 0, \dots, \kappa. \quad (19)$$

That is, $\widehat{\frac{\partial \mathbf{e}_i}{\partial \theta_{ji,m}}}(\boldsymbol{\theta}_i^k)$ can be calculated based on $\widehat{\frac{\partial \mathbf{e}_i}{\partial \theta_{ji,0}}}(\boldsymbol{\theta}_i^k)$ without additional experiments.

Combining (16)–(19), we have

$$\widehat{\frac{\partial \mathbf{e}_i}{\partial \theta_{ji,m}}}(\boldsymbol{\theta}_i^k) = \frac{\partial \mathbf{e}_i}{\partial \theta_{ji,m}}(\boldsymbol{\theta}_i^k) + \boldsymbol{\xi} \quad (20)$$

with

$$\boldsymbol{\xi} = -[(\mathbf{S} \mathbf{v}^{II})_1, \dots, (\mathbf{S} \mathbf{v}^{II})_{i-1}, (\mathbf{S} \mathbf{v}^{II})_{i+1}, \dots, (\mathbf{S} \mathbf{v}^{II})_n]^T / k_{ji}. \quad (21)$$

Since $\mathbf{v}$ is assumed to have a stochastic nature and zero mean, we have $\mathbb{E} \boldsymbol{\xi} = \mathbf{0}$ and thus

$$\mathbb{E} \widehat{\frac{\partial \mathbf{e}_i}{\partial \theta_{ji,m}}}(\boldsymbol{\theta}_i^k) = \frac{\partial \mathbf{e}_i}{\partial \theta_{ji,m}}(\boldsymbol{\theta}_i^k). \quad (22)$$

That is, the estimate of $\frac{\partial \mathbf{e}_i}{\partial \theta_{ji,m}}(\boldsymbol{\theta}_i^k)$ in (19) is unbiased. Then, from (8), the estimate of $\frac{\partial J_i}{\partial \theta_{ji,m}}(\boldsymbol{\theta}_i^k)$ can be given by

$$\widehat{\frac{\partial J_i}{\partial \theta_{ji,m}}}(\boldsymbol{\theta}_i^k) = \frac{1}{N} \sum_{t=1}^N \left(\widehat{\frac{\partial \mathbf{e}_i}{\partial \theta_{ji,m}}}(\boldsymbol{\theta}_i^k) \right)^T \mathbf{Q}_i \mathbf{e}^I. \quad (23)$$

Combining (20) and (23) yields

$$\mathbb{E} \widehat{\frac{\partial J_i}{\partial \theta_{ji,m}}}(\boldsymbol{\theta}_i^k) = \frac{\partial J_i}{\partial \theta_{ji,m}}(\boldsymbol{\theta}_i^k) + \frac{1}{N} \mathbb{E} \left[\sum_{t=1}^N \boldsymbol{\xi}^T \mathbf{Q}_i \mathbf{e}^I \right]. \quad (24)$$

Since the disturbance $\mathbf{v}$ has zero mean, and $\mathbf{v}^I$ in the tracking experiment I and $\mathbf{v}^{II}$ in the gradient experiment II are uncorrelated, it can be easily obtained that $\frac{1}{N} \mathbb{E} [\sum_{t=1}^N \boldsymbol{\xi}^T \mathbf{Q}_i \mathbf{e}^I] = 0$. Therefore, the estimate of $\frac{\partial J_i}{\partial \theta_{ji,m}}(\boldsymbol{\theta}_i^k)$ in (23) is unbiased.

In conclusion, to compute the gradient of $\mathbf{e}_i$ w.r.t. to all parameters $\boldsymbol{\theta}_i$, one tracking experiment and $(n-1)$ gradient experiments are necessary. Then, $\frac{\partial J_i}{\partial \boldsymbol{\theta}_i}(\boldsymbol{\theta}_i^k)$ can be unbiasedly estimated based on $\widehat{\frac{\partial \mathbf{e}_i}{\partial \boldsymbol{\theta}_i}}(\boldsymbol{\theta}_i^k)$ from $(n-1)$ gradient experiments and $\mathbf{e}^I$ from the tracking experiment. This is a key property of the proposed data-based IDDC approach. According to (9), $\mathbf{H}_i^k$ is estimated using $\widehat{\frac{\partial \mathbf{e}_i}{\partial \boldsymbol{\theta}_i}}(\boldsymbol{\theta}_i^k)$.

C. Data-Based IDDC Procedure

With a predesigned stabilizing $\mathbf{C}(z)$, the proposed data-based IDDC approach for the i th ($i = 1, \dots, n$) DOF is presented in the following procedure.

- 1) Set $k = 0$, and initialize $\boldsymbol{\theta}_i^0 = \mathbf{0}$.
- 2) Perform a tracking experiment and $(n-1)$ gradient experiments, and then, estimate $\frac{\partial J_i}{\partial \boldsymbol{\theta}_i}(\boldsymbol{\theta}_i^k)$ and $\mathbf{H}_i^k$.
- 3) Determine $\boldsymbol{\theta}_i^{k+1}$ according to (7).

- 4) Set $k \rightarrow k + 1$ and go back to Step 2 until an appropriate convergence condition is satisfied.

Throughout the proposed data-based IDDC procedure, only experimental data are exploited to iteratively update θ_i such that the corresponding coupling-induced error is minimized. No model nor system knowledge is needed. Furthermore, to some extent the effect of the uncertainty and machine-to-machine variation are accounted for by the measured error.

Remark 1: Although the estimated $\hat{\mathbf{H}}_i^k$ is biased, it does not affect the convergence of (7) [27]. Owing to the unbiasedness of the estimated $\frac{\partial J_i}{\partial \theta_i}(\theta_i^k)$ and the nearness of θ_i^k to θ_i^0 , a fast convergence to θ_i^* can be achieved by (7) regardless of the order κ of $D_{ij}(z)$ [36], [37].

Remark 2: The quadratic performance criterion (5) with time-weighting is used in the proposed data-based IDDC approach. The proposed approach can be easily extended toward other performance criteria, such as frequency weighting [36], that depend on the specific situation and application at hand.

IV. PERFORMANCE ASSESSMENT

In this section, performance comparisons are carried out among the following three decoupling control methods:

- 1) M_1 : Static decoupling control.
- 2) M_2 : Pre-existing data-based IDDC, see [34] for details.
- 3) M_3 : Proposed data-based IDDC.

Unlike M_3 , M_2 employs the parameter perturbation method to estimate the error gradient $\frac{\partial e_i}{\partial \theta_i}(\theta_i^k)$ for calculating $\frac{\partial J_i}{\partial \theta_i}(\theta_i^k)$ and $\mathbf{H}_i^k$

$$\frac{\partial e_i}{\partial \theta_{ji,m}}(\theta_i^k) \approx \frac{\mathbf{e}_i(\theta_i^k + \Delta\theta_{ji,m}) - \mathbf{e}_i(\theta_i^k)}{\delta\theta_{ji,m}} \quad (25)$$

with $\Delta\theta_{ji,m}$ a column with zeros except from the element corresponding to $\theta_{ji,m}$ which equals the parameter perturbation $\delta\theta_{ji,m}$. When $\mathbf{D}(z) = \mathbf{0}$, i.e., $\theta = \mathbf{0}$, M_2 and M_3 are identical to M_1 .

A. Simulation Results

In this section, a simulation example is provided to illustrate the convergence property of the proposed M_3 . Suppose that the plant is described by

$$\mathbf{P}(z) = \begin{bmatrix} \frac{0.15z^{-1}}{1-0.8z^{-1}} & \frac{0.03z^{-2}}{1-0.8z^{-1}} \\ -\frac{0.03z^{-2}}{1-0.8z^{-1}} & \frac{0.1z^{-1}}{1-0.8z^{-1}} \end{bmatrix} \quad (26)$$

which is controlled by

$$\mathbf{C}(z) = \begin{bmatrix} 2.1 + \frac{0.52}{1-z^{-1}} & 0 \\ 0 & 3.1 + \frac{0.77}{1-z^{-1}} \end{bmatrix}. \quad (27)$$

The sample interval T_s is 0.1 s and the sampling number N is 150. r_1 and r_2 are excited sequentially with a square wave

$$r(t) = \begin{cases} 1, & \text{if } 50 \leq t < 100 \\ 0, & \text{else} \end{cases}. \quad (28)$$

Assume that $\mathbf{v}$ is normally distributed white noise with zero mean and standard variance σ .

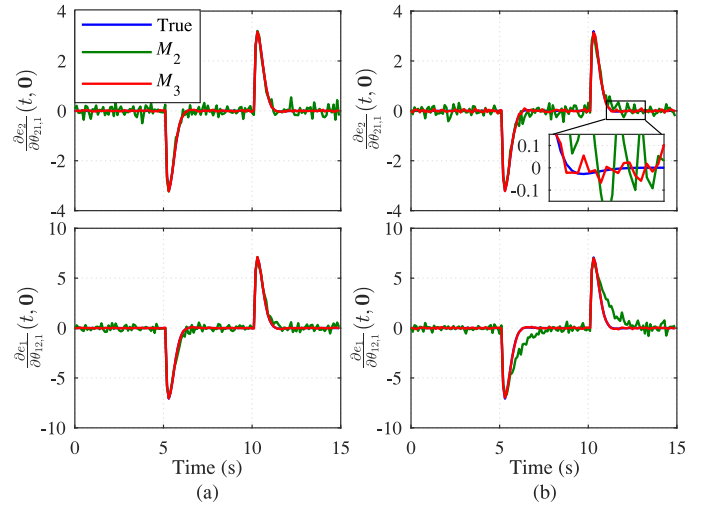


Fig. 4. Estimated $\frac{\partial e_2}{\partial \theta_{21,1}}(t, 0)$ and $\frac{\partial e_1}{\partial \theta_{12,1}}(t, 0)$ when (a) $\sigma = 0.02$ and (b) $\sigma = 0.04$.

According to (3), $D_{21}^*(z) = 0.3z^{-1}$ and $D_{12}^*(z) = -0.2z^{-1}$. Therefore, with $\kappa = 1$, $\mathbf{D}(z)$ definitely has the ability to diagonalize $\mathbf{P}(\mathbf{I} + \mathbf{D})$. For a clear presentation of simulation results, only $\theta_{21,1}$ and $\theta_{12,1}$ are selected to be iteratively tuned by M_2 and M_3 . Hence, set $D_{21}(z) = \theta_{21,1}z^{-1}$ and $D_{12}(z) = \theta_{12,1}z^{-1}$, which is a special case of (4) with $\theta_{21,0} = 0$ and $\theta_{12,0} = 0$. Accordingly, $\partial e_2 / \partial \theta_{21,0}$ and $\partial e_1 / \partial \theta_{12,0}$ are set to be zero in the parameter iterative law (7).

Fig. 4(a) and (b) shows $\frac{\partial e_2}{\partial \theta_{21,1}}(t, 0)$ and $\frac{\partial e_1}{\partial \theta_{12,1}}(t, 0)$ estimated by M_2 and M_3 when $\sigma = 0.02$ and $\sigma = 0.04$, respectively. By using the difference approximation (25), $\frac{\partial e_2}{\partial \theta_{21,1}}(t, 0)$ and $\frac{\partial e_1}{\partial \theta_{12,1}}(t, 0)$ estimated by M_2 are biased, and the involved noise is larger compared to M_3 . Furthermore, when the noise level of the closed-loop system is high, e.g., $\sigma = 0.04$, $\delta\theta_{ji,m}$ has to be selected as a relatively large value such that $(\mathbf{e}_i(\theta_i^k + \Delta\theta_{ji,m}) - \mathbf{e}_i(\theta_i^k))$ in (25) is not submerged by the noise. However, a larger $\delta\theta_{ji,m}$ leads to a larger estimate bias of $\frac{\partial e_i}{\partial \theta_{ji,m}}$, as shown in the lower part of Fig. 4(b). Compared to M_2 , M_3 provides an unbiased and precision estimate of $\frac{\partial e_2}{\partial \theta_{21,1}}(t, 0)$ and $\frac{\partial e_1}{\partial \theta_{12,1}}(t, 0)$ in the case of both $\sigma = 0.02$ and $\sigma = 0.04$.

Two hundred Monte Carlo simulations [38] are performed, where the standard variance σ of $\mathbf{v}$ conforms to the continuous uniform distribution on the interval $[0, 0.02]$. Starting from $\theta_{21,1}^0 = 0$ and $\theta_{12,1}^0 = 0$, five iterations are conducted for each DOF. Fig. 5 shows the optimized parameters $\theta^5 = [\theta_{21,1}^5, \theta_{12,1}^5]^T$ obtained via M_2 and M_3 . We also plot the resulting 95% confidence ellipse of each method, which reflects the covariance of θ^5 . The center of the confidence ellipse θ_m^5 represents the mean value of θ^5 . θ_m^5 obtained via M_2 and M_3 are $[0.3025, -0.1991]^T$ and $[0.3001, -0.2000]^T$, respectively. The former gets farther from θ^* than the latter, since $\frac{\partial e_2}{\partial \theta_{21,1}}(t, \theta_{21,1}^k)$ and $\frac{\partial e_1}{\partial \theta_{12,1}}(t, \theta_{12,1}^k)$ estimated by M_2 is biased. Furthermore, due to the larger noise involved in $\frac{\partial e_2}{\partial \theta_{21,1}}(t, \theta_{21,1}^k)$ and $\frac{\partial e_1}{\partial \theta_{12,1}}(t, \theta_{12,1}^k)$, the area of the confidence ellipse of M_2 is 2.42 times that of M_3 . Table I shows the detailed mean value and variance of $\theta_{21,1}^5$ and $\theta_{12,1}^5$, where $\text{var}(\cdot)$

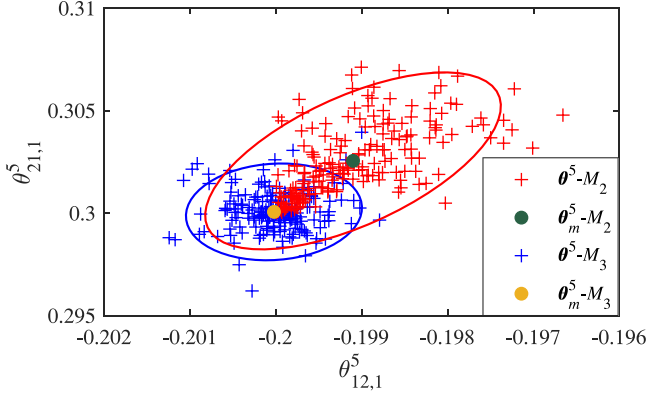

 Fig. 5. θ^5 obtained from 200 Monte Carlo simulations.

 TABLE I
MEAN VALUE AND VARIANCE OF $\theta^5_{21,1}$ AND $\theta^5_{12,1}$

	$\mathbb{E}(\theta^5_{21,1})$	$\text{var}(\theta^5_{21,1})$	$\mathbb{E}(\theta^5_{12,1})$	$\text{var}(\theta^5_{12,1})$
M_2	0.3025	3.1121×10^{-6}	-0.1991	4.9528×10^{-7}
M_3	0.3001	9.3079×10^{-7}	-0.2000	1.7547×10^{-7}

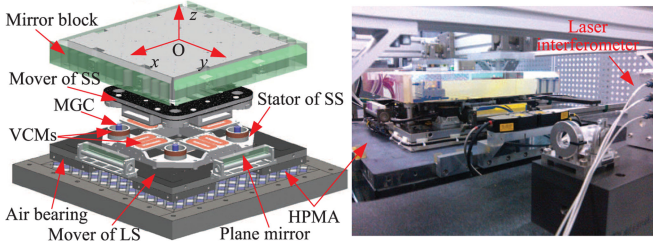


Fig. 6. Ultraprecision wafer stage—schematic and actual assembly.

denotes the variance of $\cdot$. It is clear that the estimate variance of M_3 is smaller than that of M_2 . Therefore, owing to the unbiased and precision estimate of $\frac{\partial e_2}{\partial \theta_{21,1}}(t, \theta^k_{21,1})$ and $\frac{\partial e_1}{\partial \theta_{12,1}}(t, \theta^k_{12,1})$, M_3 achieves an unbiased estimate of θ^* combined with a smaller estimate variance compared to M_2 .

B. Experimental Results

In this section, the proposed M_3 is applied to the decoupling control of a nanopositioning wafer stage developed in our lab, validating its enhanced performance in comparison with M_1 and M_2 .

1) *Experimental Setup*: The ultraprecision wafer stage developed in our laboratory is depicted in Fig. 6. In order to achieve nanoscale tracking accuracy over a long travel range, it employs a dual-stroke principle with a long-stroke stage (LS) for micropositioning and a short-stroke stage (SS) for nanopositioning. LS is a planar actuator moving along the surface of a Halbach permanent magnet array and provides support for SS along the different fields of the wafer. SS is a 6-DOF motion stage driven by voice coil motors (VCMs) and supports the wafer. The reference tracking is conducted in x and y directions so as to enable

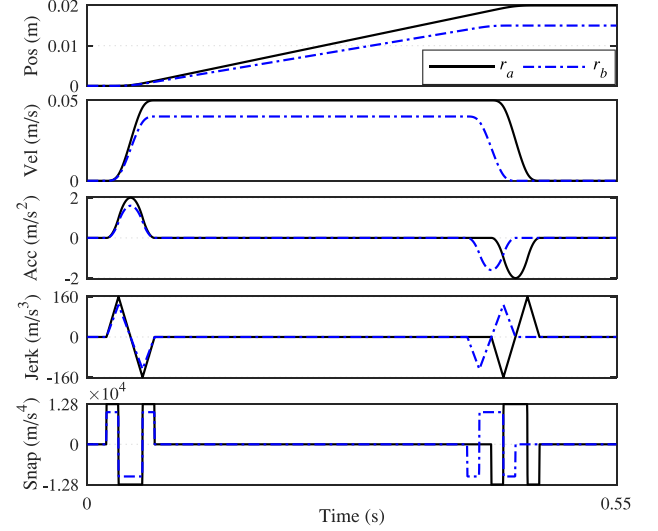


Fig. 7. Two fourth-order reference trajectories. “position,” “velocity,” and “acceleration” are abbreviated as “Pos,” “Vel,” and “Acc,” respectively. Trajectory r_a —maximum snap = 12 800 m/s⁴, maximum jerk = 160 m/s³, maximum acc = 2 m/s², cruising velocity = 0.05 m/s, and range = 0.02 m. Trajectory r_b —maximum snap = 10 240 m/s⁴, maximum jerk = 127 m/s³, maximum acc = 1.6 m/s², cruising velocity = 0.04 m/s, and range = 0.015 m.

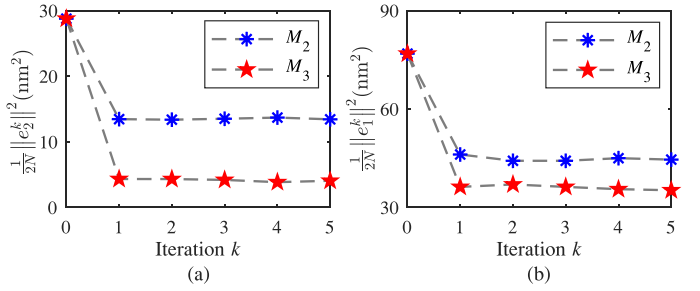


Fig. 8. Comparative results in terms of (a) $\frac{1}{2N} \|e_2^k\|^2$ when optimizing θ_{21} and (b) $\frac{1}{2N} \|e_1^k\|^2$ when optimizing θ_{12} .

full wafer coverage. The main sensor systems of the wafer stage are based on interferometry. The wafer stage is controlled in six logical axes x , y , θ_z , z , θ_x , and θ_y , and the sample interval T_s is 200 μ s.

In this paper, the decoupling control of the x -axis and y -axis of SS is considered. Therefore, the first and second DOF represent the x -axis and y -axis of SS, respectively. After a model-based static decoupling, magnitude–frequency characteristics of the x -axis and y -axis of SS are depicted in Fig. 2. Typically, two PID-type feedback controllers $C_1(z)$ and $C_2(z)$ have been pre-designed for each axis to obtain proper stability margin and sensitivity magnitude for disturbance rejection, see also [9], [30]. However, the residual coupling limits the further improvement of the servo performance.

Since there are only two DOFs considered in the DDC design, it follows that $\theta_1 = \theta_{21}$, $\theta_2 = \theta_{12}$, $e_1 = e_2$, and $e_2 = e_1$ (see Section III-A). The weighting factor in (5) is set as one for each DOF. Consequently, $J_1(\theta_1) = \frac{1}{2N} \mathbb{E} \|e_2\|^2$ and $J_2(\theta_2) = \frac{1}{2N} \mathbb{E} \|e_1\|^2$. Two fourth order reference trajectories with length $N = 2750$ are considered as depicted in Fig. 7.

TABLE II
PERFORMANCE INDICES OF M_2 AND M_3 AT ITERATION 5
IN COMPARISON WITH M_1

		M_1	M_2	M_3
$\mathbf{r} = [r_a, 0]^T$	$\frac{1}{2N} \ e_2^5\ ^2 \text{ (nm}^2\text{)}$	28.8	13.4	4.1
	$\max(e_2^5) \text{ (nm)}$	30	23	14
$\mathbf{r} = [0, r_a]^T$	$\frac{1}{2N} \ e_1^5\ ^2 \text{ (nm}^2\text{)}$	76.9	44.6	35.2
	$\max(e_1^5) \text{ (nm)}$	48	38	38

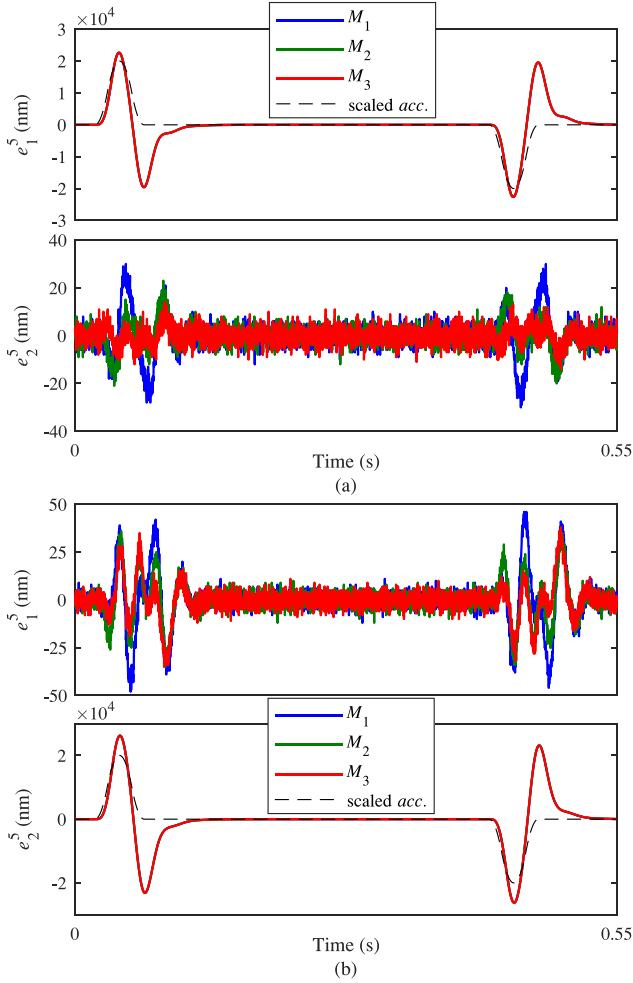


Fig. 9. Time-domain error signals e_1^5 and e_2^5 during the iterative optimization of (a) θ_{21} with $\mathbf{r} = [r_a, 0]^T$ and (b) θ_{12} with $\mathbf{r} = [0, r_a]^T$.

Trajectory r_a is used to sequentially excite each DOF, and $\mathbf{D}(z)$ is iteratively optimized DOF-by-DOF via M_2 and M_3 . Then, the effectiveness of $\mathbf{D}(z)$ optimized under r_a is validated again in case that these two DOFs are sequentially excited by r_b .

2) Experimental Results: Starting from $\theta_i^0 = \mathbf{0}$ ($i = 1, 2$), five iterations are conducted for each DOF. Fig. 8 shows comparative results in terms of $\frac{1}{2N} \|e_2^k\|^2$ when optimizing θ_{21} and $\frac{1}{2N} \|e_1^k\|^2$ when optimizing θ_{12} . Table II details the performance indices of M_2 and M_3 at iteration 5 in comparison with M_1 . As shown in Fig. 8, the first iteration associates with the main

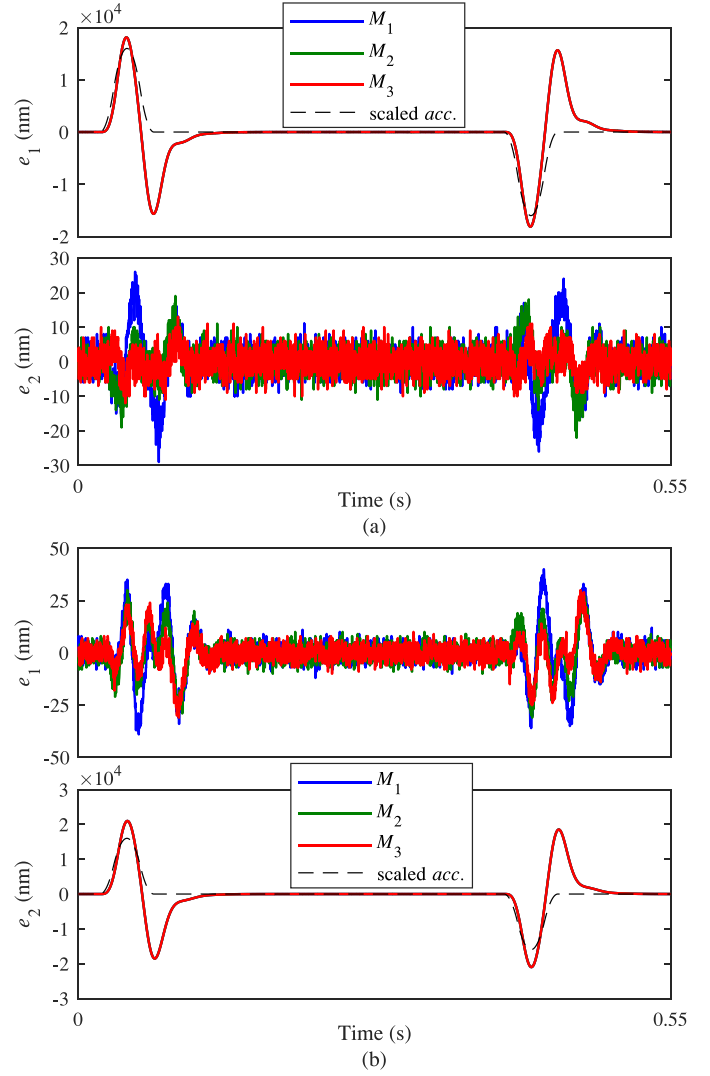


Fig. 10. Time-domain error signals with $\theta = [(\theta_1^5)^T, (\theta_2^5)^T]^T$ obtained from the iterative optimization under r_a when (a) $\mathbf{r} = [r_b, 0]^T$ and (b) $\mathbf{r} = [0, r_b]^T$.

improvement. This favors the fast convergence of (7) as θ_i^* are close to zero. Notice that $\theta_i^0 = \mathbf{0}$ corresponds to the state where M_1 is employed in the closed-loop system. It is clear that both M_2 and M_3 outperform M_1 due to the data-based dynamic decoupling. $\frac{1}{2N} \|e_2^5\|^2$, when optimizing θ_{21} for M_2 and M_3 , are 13.4 and 4.1 nm², respectively. $\frac{1}{2N} \|e_1^5\|^2$, when optimizing θ_{12} for M_2 and M_3 , are 44.6 and 35.2 nm², respectively. Therefore, servo performance is significantly improved by M_3 compared to M_2 . This conclusion is also confirmed by the time-domain error signals shown in Fig. 9, where the dash curve represents the scaled acceleration set points. For example, when optimizing θ_{21} , the peak value of $|e_2^5|$ is decreased from 30 nm (corresponding to M_1) to 23 nm by M_2 , and to 14 nm by M_3 .

In order to validate the robustness of M_3 , performance comparisons are further conducted by employing another trajectory r_b to sequentially excite each DOF. Herein, $\mathbf{D}(z)$, optimized under r_a , is used for M_2 and M_3 . Fig. 10 shows that the

TABLE III
PERFORMANCE INDICES WHEN r_b SEQUENTIALLY EXCITES EACH DOF

		M_1	M_2	M_3
$\mathbf{r} = [r_b, 0]^T$	$\frac{1}{2N} \ e_2\ ^2$ (nm ²)	20.7	11.0	6.5
	$\max(e_2)$ (nm)	29	22	13
$\mathbf{r} = [0, r_b]^T$	$\frac{1}{2N} \ e_1\ ^2$ (nm ²)	52.5	31.3	24.8
	$\max(e_1)$ (nm)	40	30	31

coupling-induced error is significantly decreased by M_2 and M_3 compared to M_1 , and M_3 performs better than M_2 . Table III presents the detailed performance indices. When $\mathbf{r} = [r_b, 0]^T$, $\frac{1}{2N} \|e_2\|^2$ for M_1 , M_2 , and M_3 are 20.7, 11.0, and 6.5 nm², respectively. When $\mathbf{r} = [0, r_b]^T$, $\frac{1}{2N} \|e_1\|^2$ for M_1 , M_2 , and M_3 are 52.5, 31.3, and 24.8 nm², respectively. These results further illustrate the enhanced performance of M_3 in comparison with M_1 and M_2 .

V. CONCLUSION

In this paper, to minimize the internal coupling and improve the servo performance, a data-based IDDC approach was proposed for precision MIMO motion systems. Atop a static decoupling part, a MIMO DDC controller structured with FIR filter was considered to deal with the residual coupling. With the aim of minimizing the coupling-induced error, gradient-based GNA was typically utilized to iteratively tune the controller parameters. Accordingly, an unbiased estimate method for the gradient of the performance criterion was developed, based entirely on I/O data of the closed-loop MIMO system. Owing to the unbiasedness of the estimated gradient and the nearness of the optimal parameters to zero, a fast convergence can be achieved by the proposed approach regardless of the controller order. Comparative simulation results confirmed that the proposed approach achieved the unbiased estimate of the optimal parameters combined with a small estimate variance. Application to an ultraprecision wafer stage validated the enhanced performance of the proposed approach in comparison with pre-existing approaches.

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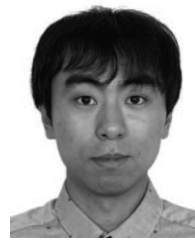
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